

NutriCart AI

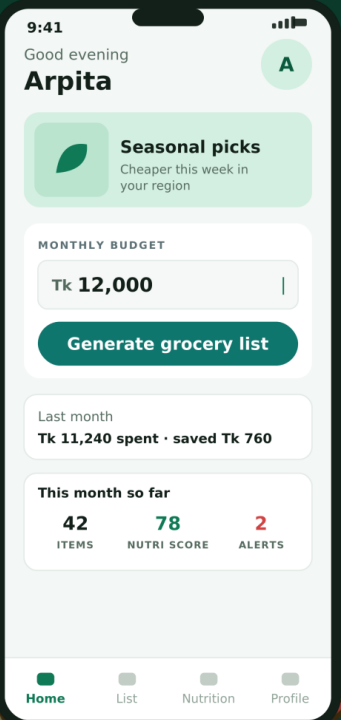
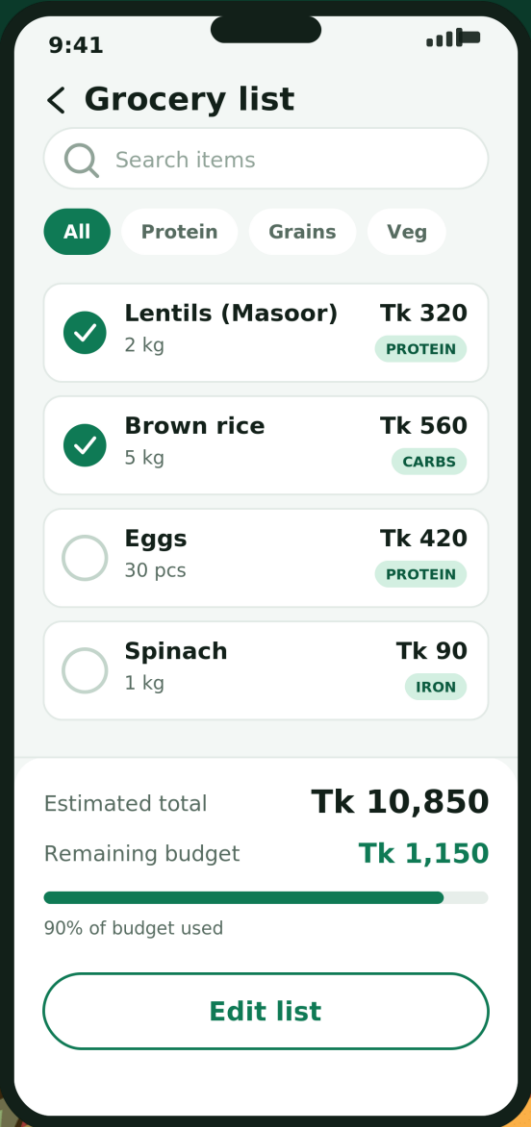
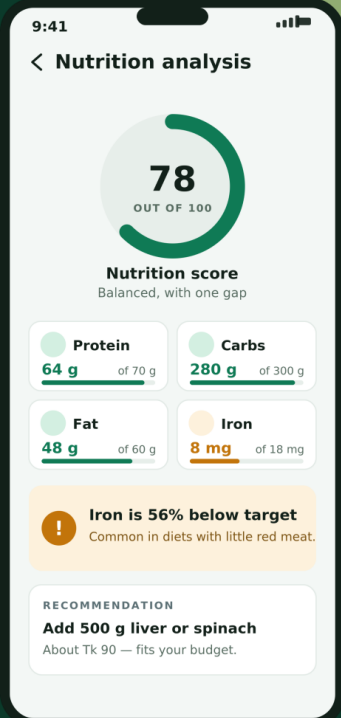
Screen Details

A budget- and nutrition-aware grocery list generator. Nine screens specified to Jetpack Compose level — every element numbered, measured in dp and sp, named against a design token, and paired with the behaviour it triggers.

NAME	EXAM ROLL	ROLL	PLATFORM
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Mobile Application Development (CSE-410) · 4th Year 1st Semester
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Budget in, balanced list out — checked against allergies and health conditions.



Six conventions hold across all fifteen screen slides.

1 Numbered badges

Every callout on a mockup carries a number. The row with the same number in the table beside it is that element's full specification.

2 Colours are named, not hard-coded

Specs reference tokens such as `primary` or `warning.container`. Slide 3 is the only place a hex value appears — resolve every token there and nowhere else.

3 Units

dp for anything measured, sp for anything typed, ms for motion. No pixel values appear in this document.

4 Components are specified once

Slide 5 defines thirteen reusable components. A row that reads C-03 inherits that entire specification; only the differences are restated.

5 Screen IDs are permanent

SCR-01 to SCR-09 are used again in the UI Flow document, so a node there points at exactly one slide here.

6 Footer strip

Purpose, where the screen is entered from, where it exits to, which data feeds it, and every state it must handle.



The only slide in this document that contains a hex value. Every screen specification names a token and resolves it here.

Token	Light	Dark	Used for
surface	#F5F8F6	#0E1A15	Screen background
surface.card	#FFFFFF	#16241E	Cards, list rows, sheets
surface.sunken	#E7EEEA	#0A1310	Progress rails, stepper buttons, input wells
on.surface	#122019	#E9F2EC	Item names, headings, values
on.surface.muted	#566B60	#9CB3A7	Labels, units, helper text
on.surface.faint	#90A398	#6D857A	Placeholders, unselected nav
primary	#0F7A55	#3BD495	Filled buttons, checked states, on-target bars
on.primary	#FFFFFF	#04231A	Text and icons on primary
primary.container	#D3EFE1	#0E3B2B	Nutrition tags, chips, promo banner
on.primary.container	#0A5A3E	#A5E9C7	Text on primary.container
warning	#C2740A	#F0A83C	Deficiency alerts, health-condition conflicts
warning.container	#FDF1DC	#33260E	Warning card background
danger	#D14343	#F08585	Allergy conflicts, delete, log out
danger.container	#FCECEC	#331E1E	Allergy card background
outline	#E2EAE6	#24352E	Card hairlines and dividers

Dark values are tuned for contrast, not inverted. primary lifts to 3BD495 so a budget rail stays legible on a dark surface; warning and danger lift for the same reason.

Nine type roles and nine layout rules. No value outside these appears on any screen slide.

Role	Size	Weight	Line	Used for
display	32sp	Bold	38sp	Splash wordmark, user name
headline	26sp	Bold	32sp	App bar titles, screen titles
title.lg	22sp	Bold	28sp	Card headings, section titles
title	20sp	Bold	26sp	Item names, button labels
stat	32sp	Bold	36sp	Budget figures, nutrition score, totals
body	16sp	Regular	22sp	Descriptions, warning text
caption	14sp	Regular	19sp	Units, quantities, helper lines
label	12sp	Bold	16sp	ALL-CAPS section labels, +1.5sp tracking
micro	11sp	Bold	14sp	Nutrition tags, alert badges, nav labels

Layout rules

Design viewport 393 x 832 dp (Pixel-class); all values scale from this

Screen gutter 20dp left and right on every screen

Spacing scale 4 / 8 / 12 / 16 / 20 / 24 / 32 dp — no other values

Corner radius input 10dp · card 14dp · sheet 22dp · button and chip 50%

Elevation Cards 0dp + 1dp outline · docked total bar 6dp · dialog 12dp

Touch target 48 x 48dp minimum; stepper and checkbox hit areas are padded to it

Icon sizes 20dp inline · 24dp nav and app bar · 40dp alert and empty-state

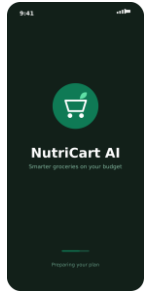
Motion Bars and rings 400ms · sheet 250ms · nav fade 150ms, standard easing

Type face Roboto — the Compose default; no font is bundled with the APK

Thirteen reusable components. A screen row that reads C-03 inherits this entire specification; only differences are restated.

ID	Component	Composable	Specification	Appears on
C-01	Filled button	Button	56dp tall, full width minus gutters, pill, primary fill, title/20sp	SCR-02,03,04,06,08
C-02	Outlined button	OutlinedButton	52dp, pill, 2dp border, label in the border colour; danger variant for log out	SCR-05, 08, 09
C-03	Card	Card	14dp radius, surface.card, 1dp outline, 0dp elevation, 16dp inner padding	All screens
C-04	Input field	OutlinedTextField	50dp, 10dp radius, 1dp outline, 2dp primary when focused, title/20sp	SCR-04, 05
C-05	Dropdown	ExposedDropDownMenuBox	50dp, 10dp radius, trailing 20dp chevron, searchable, required	SCR-03
C-06	Stepper	Row + 2 IconButton	36dp circular buttons on surface.sunken, value in stat/24sp between them	SCR-03, 06
C-07	Checkbox	Checkbox	22dp box, 6dp radius, primary when checked, 48dp hit area, multi-select	SCR-03, 05
C-08	Radio	RadioButton	22dp, primary ring when selected, single-select, exactly one required	SCR-03
C-09	Tag chip	AssistChip	32dp, pill; primary.container for nutrients, danger for allergies, warning for conditions	SCR-05, 08, 09
C-10	Section label	Text	label/12sp caps, +1.5sp tracking, on.surface.muted, 12dp below the block above	All screens
C-11	Progress rail	LinearProgressIndicator	8dp tall, pill, primary on surface.sunken; warning below 60% of target	SCR-05, 06, 07
C-12	Bottom nav	NavigationBar	72dp, 4 items (Home, List, Nutrition, Profile), primary when selected	SCR-04, 09
C-13	Alert card	Card + Row	14dp radius, 6dp leading colour bar, 28dp icon circle, tinted container background	SCR-07, 08

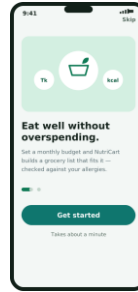
Nine screens. These IDs are reused unchanged in the UI Flow document.



SCR-01

Splash

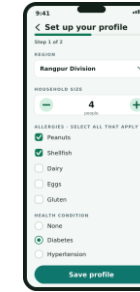
Entry



SCR-02

Welcome / Onboarding

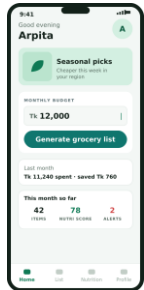
Entry



SCR-03

User profile setup

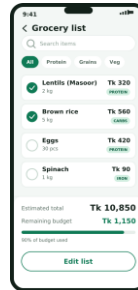
Entry



SCR-04

Home dashboard

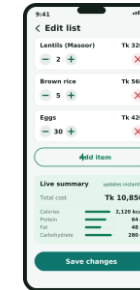
Daily use



SCR-05

Generated grocery list

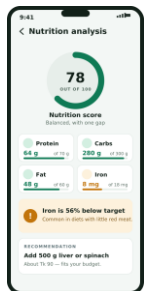
Grocery



SCR-06

Edit grocery list

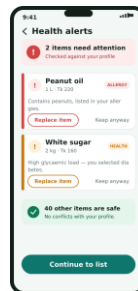
Grocery



SCR-07

Nutrition analysis

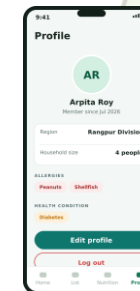
Insight



SCR-08

Allergy and health alerts

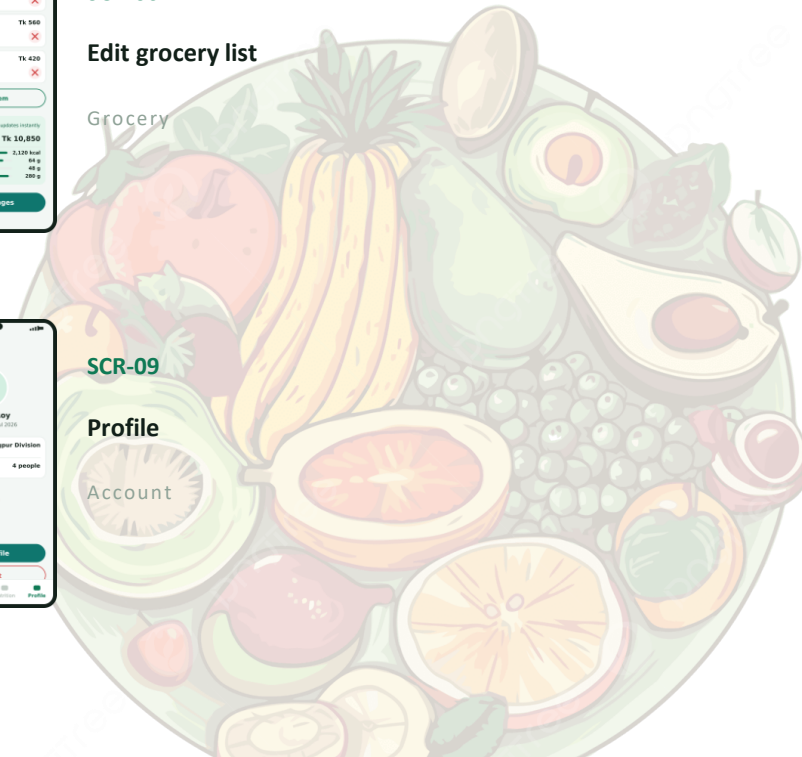
Insight



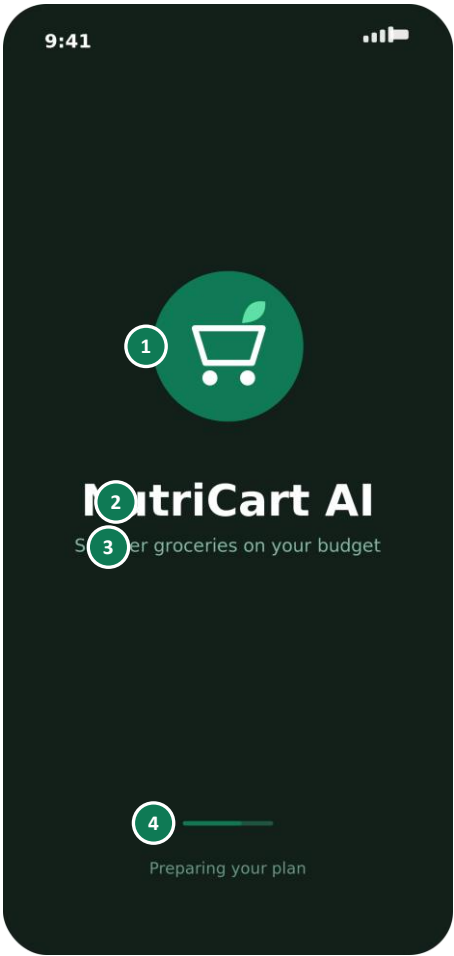
SCR-09

Profile

Account



Holds the first frame while the profile and cached list are read from local storage.



#	Element	Composable	Specification	Behaviour
1	App logo	Canvas	200dp circle, 4dp primary ring, cart glyph with a leaf mark	Scales 0.92 to 1.0 over 220ms
2	App name	Text	display/32sp Bold, on.primary	Static
3	Tagline	Text	caption/14sp, primary at 70% alpha, centred	Static
4	Loading indicator	LinearProgressIndicator	80 x 4dp, pill, primary on a 20% track	Indeterminate; hidden if ready under 400ms
5	Background	Surface	Token on.surface in both themes — splash is always dark	Does not follow the theme setting
6	Route decision	LaunchedEffect	Minimum 600ms, maximum 3000ms on screen	Reads the profile row, then navigates
7	Visibility	—	Not in the back stack; cannot be returned to	popUpTo(inclusive = true) on exit
8	Failure path	Snackbar on SCR-02	If storage throws, onboarding opens and surfaces a retry	Never blocks on the splash

PURPOSE

Holds the first frame while the profile and cached list are read from local storage.

ENTERED FROM

Cold launch (launcher icon)

EXITS TO

SCR-02 first run · SCR-04 once a profile exists

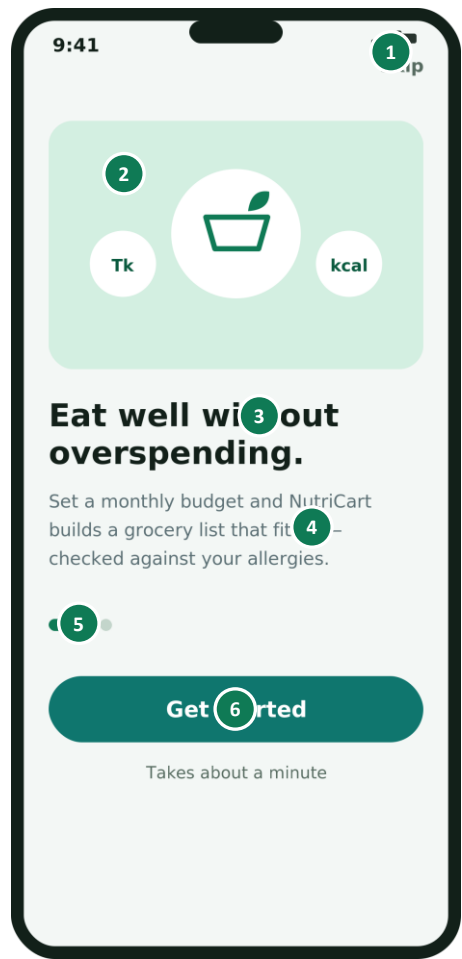
DATA SOURCE

ProfileDao.exists() · SettingsStore.onboardingComplete

STATES

Loading · migration · corrupt-store fallback

Introduces the value in one sentence and sends the user into profile setup.



#	Element	Composable	Specification	Behaviour
1	Skip button	TextButton	caption/14sp Bold, on.surface.muted, 20dp gutter, 48dp target	Marks onboarding done, jumps to SCR-04 with an unset profile
2	Illustration banner	Box + Canvas	Full width minus gutters, 216dp tall, 20dp radius, primary.container fill	Decorative; contentDescription is null
3	Welcome heading	Text	headline/26sp Bold, on.surface, max 2 lines, 20dp gutter	Changes per pane
4	Description text	Text	body/16sp, on.surface.muted, 24sp line height, max 3 lines	Changes per pane
5	Pager dots	Row + Box	8dp dots, active 28 x 8dp pill primary, 10dp gap	Reflects HorizontalPager state
6	Get started	C-01	56dp, primary, 20dp gutters	Advances the pager; the last pane opens SCR-03
7	Footnote	Text	caption/14sp on.surface.muted, centred	Sets the expected time cost before setup
8	Pager	HorizontalPager	3 pages, swipe both directions	Back on pane 1 exits the app
9	Background	Surface	Token surface	Follows the theme setting from this screen onward

PURPOSE

Introduces the value in one sentence and sends the user into profile setup.

ENTERED FROM

SCR-01 when no profile row exists

EXITS TO

SCR-03 via Get started · SCR-04 via Skip

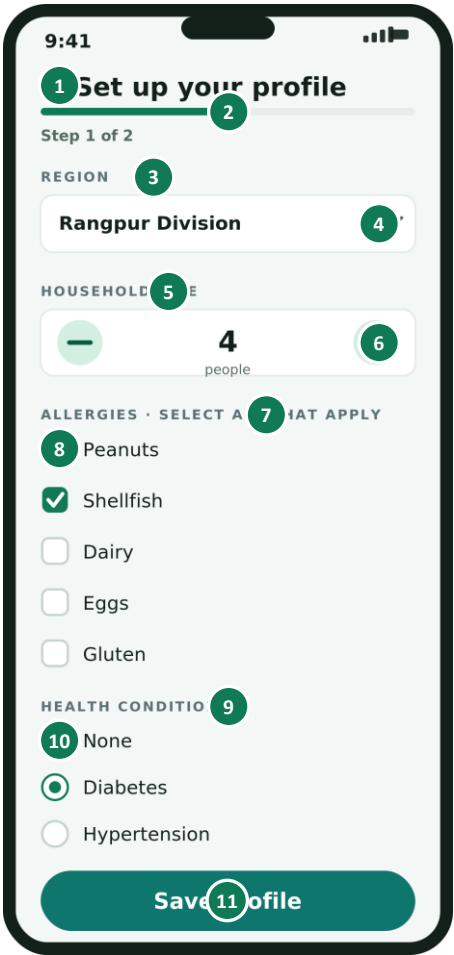
DATA SOURCE

SettingsStore.onboardingComplete (write on exit)

STATES

Pane 1 of 3 · swipe · skip

Collects the one-time information every later screen depends on: region, household size, allergies and health condition.



#	Element	Composable	Specification	Behaviour
1	App bar	TopAppBar	56dp, back chevron 24dp, headline/26sp Bold title	Back warns before discarding entered values
2	Step indicator	LinearProgressIndicator	Full width minus gutters, 6dp, pill, primary on surface.sunken	Two steps; this is step 1
3	Region label	C-10	label/12sp caps, on.surface.muted	Static
4	Region dropdown	C-05	50dp, 10dp radius, searchable, required; default is the last saved region	Empty on submit shows an outline in danger and a message below
5	Household label	C-10	label/12sp caps, 24dp above	Static
6	Household stepper	C-06	Min 1, max 12, default 4; buttons disable at the bounds	Drives every quantity the generator produces
7	Allergy label	C-10	label/12sp caps; helper reads select all that apply	Static

PURPOSE

Collects the one-time information every later screen depends on: region, household size, allergies and health condition.

ENTERED FROM

SCR-02 · Edit profile on SCR-09

EXITS TO

SCR-04 on save

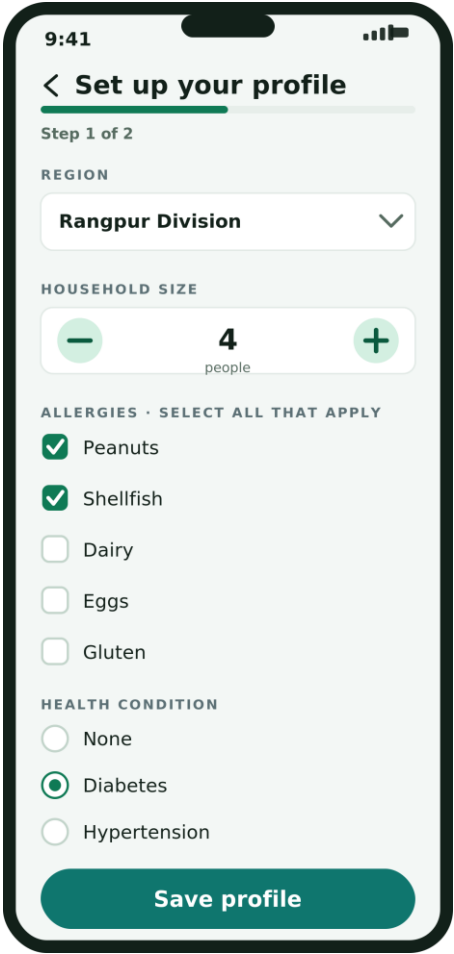
DATA SOURCE

ProfileEntity(region, householdSize, allergies, condition)

STATES

Empty · partially filled · validation error · editing an existing profile

Specification continued — same screen, same numbering.



#	Element	Composable	Specification	Behaviour
8	Allergy checkbox	C-07	22dp box, 6dp radius, primary when checked, 48dp row target	Multi-select, optional; drives the SCR-08 allergy check
9	Condition label	C-10	label/12sp caps, 24dp above	Static
10	Health condition	C-08	22dp radio, primary ring when selected, one option per row	Single-select, required; None is a valid answer
11	Save profile	C-01	56dp, primary, 20dp gutters, docked above the inset	Disabled until region and condition are both set
12	Validation	ViewModel	Region required · condition required · household 1 to 12	Errors resolve on the field, not in a dialog
13	Persistence	ProfileDao.upsert	One row; re-entering from SCR-09 loads and overwrites it	Write is atomic — no partial profile is ever stored

PURPOSE

Collects the one-time information every later screen depends on: region, household size, allergies and health condition.

ENTERED FROM

SCR-02 · Edit profile on SCR-09

EXITS TO

SCR-04 on save

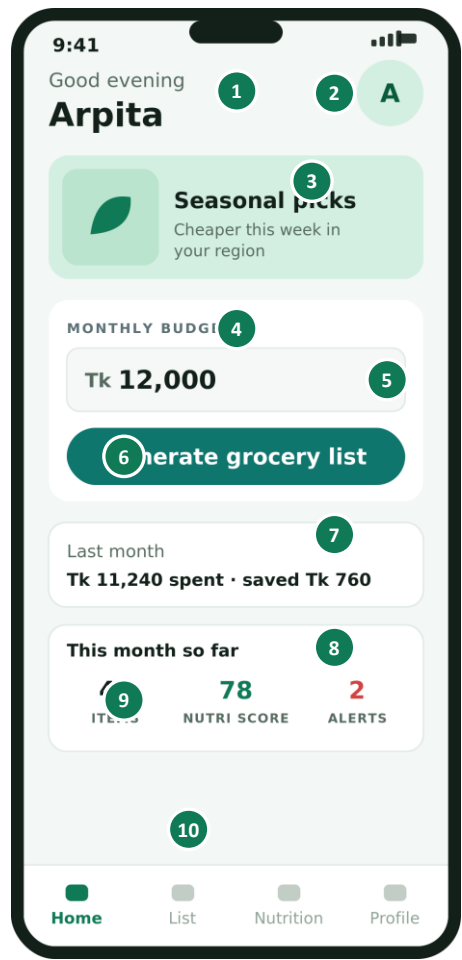
DATA SOURCE

ProfileEntity(region, householdSize, allergies, condition)

STATES

Empty · partially filled · validation error · editing an existing profile

Takes a monthly budget and turns it into a grocery list. The one number the whole app runs on.



#	Element	Composable	Specification	Behaviour
1	Greeting text	Text	caption/14sp muted over display/32sp Bold on.surface, left aligned	Morning / afternoon / evening by clock; name from the profile
2	User avatar	Box + Text	88dp circle, primary.container, initials in headline/26sp	Tap opens SCR-09
3	Promotional banner	Card + Row	Full width minus gutters, 108dp, 18dp radius, primary.container; 84dp image block at 14dp radius, cropped centre	Tap opens seasonal picks; hidden when the region has no offers
4	Budget label	C-10	label/12sp caps, on.surface.muted, inside the card	Static
5	Budget input	C-04	50dp, 10dp radius, numeric keyboard, Tk prefix, max 7 digits	Rejects non-numeric; error below 500 or above 999999
6	Generate button	C-01	50dp, primary, full card width minus padding	Disabled while blank; shows a spinner during generation
7	Previous budget card	C-03	72dp, 14dp radius, 1dp outline, spent and saved figures	Hidden entirely on first run — no zero state

PURPOSE

Takes a monthly budget and turns it into a grocery list. The one number the whole app runs on.

ENTERED FROM

SCR-01 · SCR-03 on save · bottom nav

EXITS TO

SCR-05 on generate · nav tabs

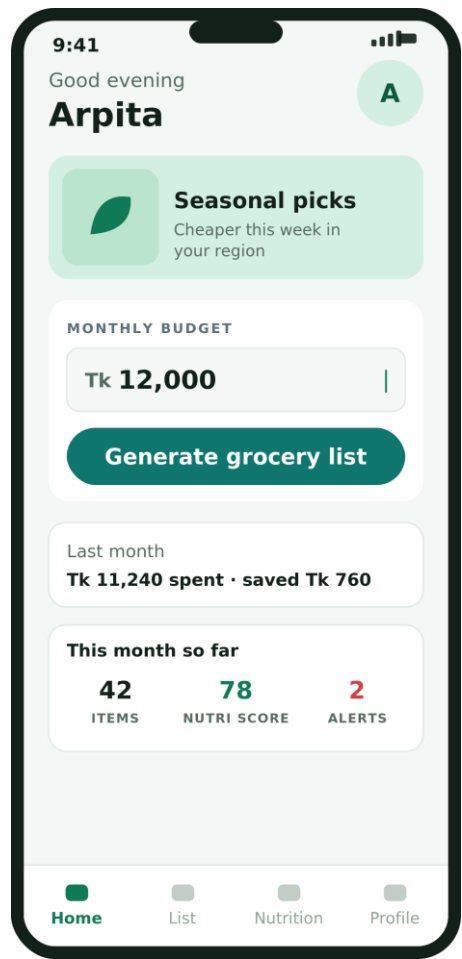
DATA SOURCE

ProfileEntity · BudgetDao.latest() · summary aggregates

STATES

Populated · no previous budget · generating · generation failed

Specification continued — same screen, same numbering.



#	Element	Composable	Specification	Behaviour
8	Quick summary card	C-03	110dp, three equal cells, stat/32sp values with label/12sp caps	Alert count uses danger; nutrition score uses primary
9	Summary cell	Column	Equal weight thirds, values centred, labels centred beneath	Tap on the alerts cell opens SCR-08
10	Bottom navigation	C-12	72dp, 4 items, primary when selected, on.surface.faint when not	Home selected; tab change is a fade
11	Scroll	Column + verticalScroll	Content scrolls under the nav bar; nav is fixed	Bottom inset padding equal to the nav height
12	Generation	GroceryGenerator	Budget, household size, region prices and the profile go in	On failure the button returns and a snackbar offers retry
13	State restore	SavedStateHandle	A typed budget survives rotation and process death	Never silently cleared

PURPOSE

Takes a monthly budget and turns it into a grocery list. The one number the whole app runs on.

ENTERED FROM

SCR-01 · SCR-03 on save · bottom nav

EXITS TO

SCR-05 on generate · nav tabs

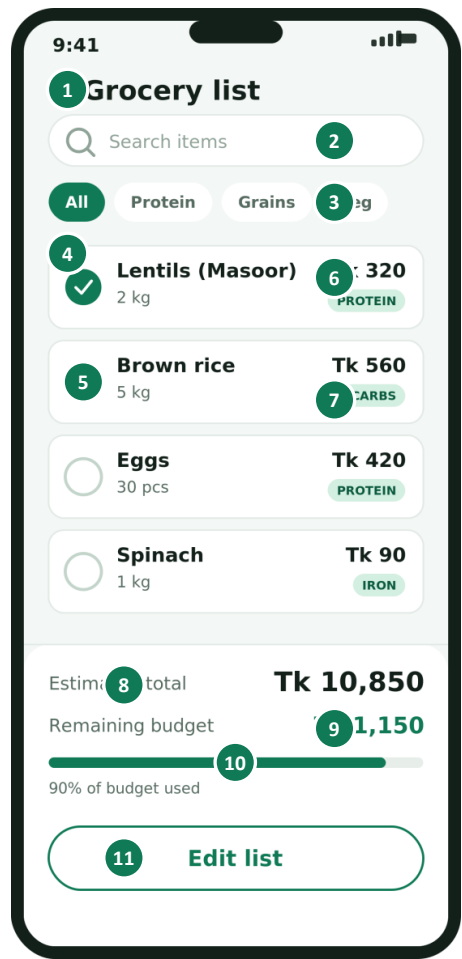
DATA SOURCE

ProfileEntity · BudgetDao.latest() · summary aggregates

STATES

Populated · no previous budget · generating · generation failed

The generated list, with running totals kept visible so the budget is never a surprise at the end.



#	Element	Composable	Specification	Behaviour
1	App bar	TopAppBar	56dp, back chevron, headline/26sp Bold title	Back returns to SCR-04; the list is already saved
2	Search bar	C-04 (pill)	44dp, pill, 20dp leading search icon, placeholder in on.surface.faint	Filters as you type; no submit action
3	Category filter	LazyRow of C-09	36dp chips, 12dp gap, primary fill when selected	Single-select; All resets the filter
4	Grocery item card	C-03	76dp, 14dp radius, 1dp outline, 12dp between cards	Swipe left reveals delete; swipe right marks bought
5	Item checkbox	C-07	32dp circle, primary fill with a check when ticked, 48dp target	Ticking dims the row; it does not remove it from the total
6	Quantity and price	Column	Quantity caption/14sp muted, price title/20sp Bold on.surface right aligned	Price recalculates when quantity changes on SCR-06
7	Nutrition tag	C-09	24dp pill, primary.container, micro/11sp Bold caps	One tag per item — its dominant macro or micronutrient

PURPOSE

The generated list, with running totals kept visible so the budget is never a surprise at the end.

ENTERED FROM

SCR-04 on generate · SCR-06 on save

EXITS TO

SCR-06 edit · SCR-07 nutrition · SCR-08 alerts

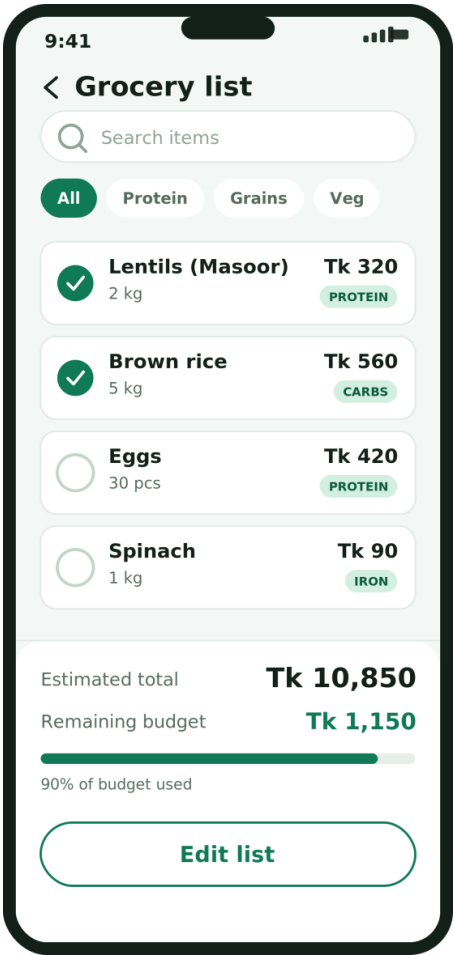
DATA SOURCE

GroceryListDao.observe(listId) + ItemDao + price table

STATES

Populated · empty result · over budget · offline prices stale

Specification continued — same screen, same numbering.



#	Element	Composable	Specification	Behaviour
8	Total cost card	Card (docked)	Docked bottom sheet, 22dp top radius, 6dp elevation	Recalculates on every quantity or tick change
9	Remaining budget	Row	caption/14sp label, stat/24sp value in primary; danger when negative	Negative values are shown, never clamped to zero
10	Budget rail	C-11	8dp, pill, primary; switches to danger past 100%	Animates over 400ms on change
11	Edit button	C-02	52dp, pill, 2dp primary border	Opens SCR-06
12	List	LazyColumn	20dp gutters, bottom padding equal to the docked card height	Content never sits under the total card
13	Empty result	Empty state	Shown when the budget cannot cover a balanced list	Suggests raising the budget by a named amount

PURPOSE

The generated list, with running totals kept visible so the budget is never a surprise at the end.

ENTERED FROM

SCR-04 on generate · SCR-06 on save

EXITS TO

SCR-06 edit · SCR-07 nutrition · SCR-08 alerts

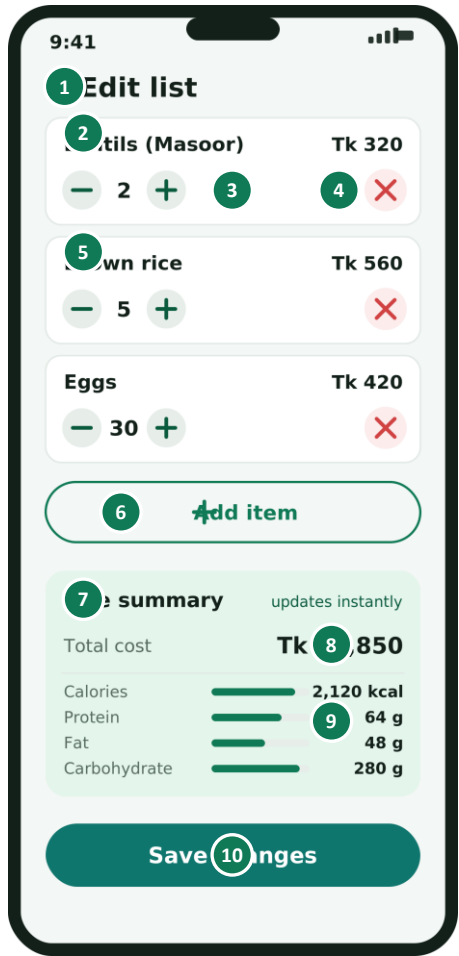
DATA SOURCE

GroceryListDao.observe(listId) + ItemDao + price table

STATES

Populated · empty result · over budget · offline prices stale

Manual correction, with cost and macros recomputed on every keystroke rather than on save.



#	Element	Composable	Specification	Behaviour
1	App bar	TopAppBar	56dp, back chevron, headline/26sp Bold title	Back warns when changes are unsaved
2	Grocery card	C-03	94dp, 14dp radius, 1dp outline — taller than SCR-05 to hold the stepper	Name and price are read-only here
3	Quantity stepper	C-06	36dp buttons, min 1, max 99, value in title/20sp between them	Every press recomputes cost and macros immediately
4	Delete button	IconButton	36dp circle, danger.container fill, danger glyph, 48dp target	Removes the row with a 5-second undo snackbar
5	Card list	LazyColumn	12dp between cards, 20dp gutters	Reorders are not supported at v1
6	Add item button	C-02	52dp, pill, 2dp primary border, leading 20dp plus icon	Opens a searchable item picker sheet
7	Live summary card	C-03	198dp, 16dp radius, primary.container tint	Recomputes on every change — never on save
8	Total cost	Row	caption/14sp label, stat/24sp Bold value right aligned	Danger token once the budget is exceeded
9	Macro rows	Row + C-11	Four rows: calories, protein, fat, carbohydrate; 86dp rail each	Rail is the value over the household daily target
10	Save changes	C-01	56dp, primary, docked above the inset	Writes the batch in one transaction, then pops to SCR-05
11	Update type	StateFlow	Real-time; the calculator runs on the ViewModel, not in composition	Debounce 120ms so holding the stepper stays smooth

PURPOSE
Manual correction, with cost and macros recomputed on every keystroke rather than on save.

ENTERED FROM
SCR-05 edit button

EXITS TO
SCR-05 on save · discard returns unchanged

DATA SOURCE
ItemDao.update / delete / insert · NutritionCalculator

STATES
Editing · unsaved changes · item limit reached

Scores the list against the household's needs and names the single largest gap.



#	Element	Composable	Specification	Behaviour
1	App bar	TopAppBar	56dp, back chevron, headline/26sp Bold title	Back returns to the caller
2	Circular progress	Canvas	148dp outer, 17dp stroke, primary sweep on a surface.sunken track	Sweeps from 0 over 400ms on entry
3	Nutrition score	Text	stat/32sp Bold centred inside the ring, label/12sp caps beneath	0 to 100; colour follows the band, not the value
4	Nutrient card	C-03	80dp, 2-column grid, 14dp radius, 1dp outline, 12dp gutter	Four cards: protein, carbohydrate, fat and the weakest micronutrient
5	Nutrient rail	C-11	8dp, pill; primary at or above 60% of target, warning below	Value over required, capped at 100% width
6	Nutrient values	Row	Current in title/20sp Bold, required in caption/14sp muted	Units come from the RDA table, never hard-coded
7	Deficiency alert	C-13	98dp, warning.container, 30dp warning icon circle	Shown only for the largest gap — never a list of warnings
8	Recommendation card	C-03	104dp, 14dp radius, 1dp outline	Names a concrete item and its cost against the remaining budget
9	Recommendation cost	Text	caption/14sp on.surface.muted	Suppressed when the item would exceed the remaining budget
10	Scoring	NutritionCalculator	Weighted distance from the RDA set for the household size	Runs on device; no network call
11	All targets met	Success state	Replaces the deficiency card with a primary.container confirmation	Score above 90 with no nutrient under 60%

PURPOSE

Scores the list against the household's needs and names the single largest gap.

ENTERED FROM

SCR-05 · bottom nav

EXITS TO

SCR-06 to act on a recommendation

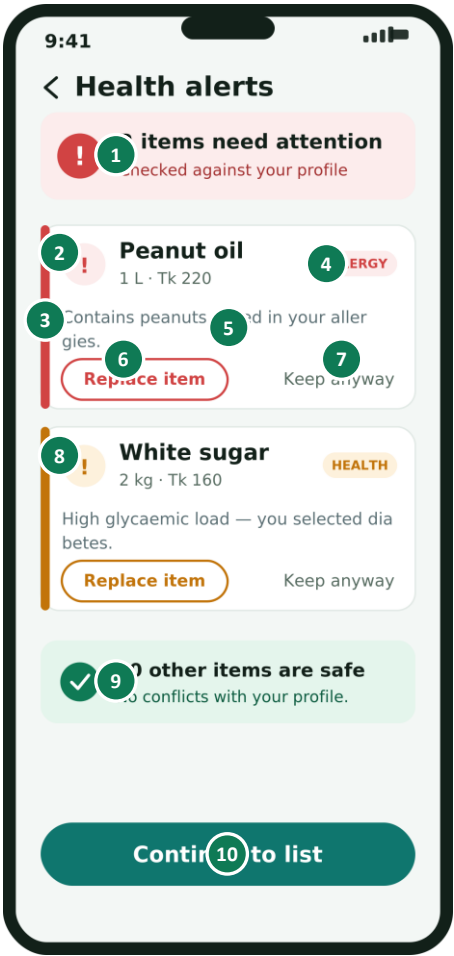
DATA SOURCE

NutritionCalculator over the list + RDA table keyed by household size

STATES

Scored · insufficient data · all targets met

Cross-checks the generated list against the profile, and lets the user act without leaving the screen.



#	Element	Composable	Specification	Behaviour
1	Alert banner	C-13	72dp, danger.container, 30dp icon circle, count in title.lg/22sp	Hidden entirely when nothing conflicts
2	Warning card	C-03	152dp, 14dp radius, one card per conflicting item	Ordered allergy first, then health condition
3	Severity bar	Box	6dp wide, full card height, leading edge; danger or warning	The only place a colour bar is used in this app
4	Alert type badge	C-09	22dp pill, tinted container, micro/11sp Bold caps	ALLERGY uses danger, HEALTH uses warning
5	Warning description	Text	body/16sp on.surface, max 2 lines	States the matched rule, not a generic caution
6	Replace item button	C-02	34dp, pill, 2dp border in the card's severity colour	Opens a picker filtered to safe substitutes at a similar price
7	Keep anyway	TextButton	caption/14sp on.surface.muted, 48dp target	Records an override on that item; the alert does not return
8	Second card	C-03	Identical metrics; the list is never truncated	Every conflict gets a card
9	Safe confirmation	C-13 (positive)	70dp, primary.container, 26dp check circle	Names how many items passed — reassurance, not silence
10	Continue button	C-01	56dp, primary, docked above the inset	Returns to SCR-05 with any replacements applied
11	Matching	AllergyMatcher	Ingredient tags against profile.allergies; rule set for the condition	Runs on device against the local item table
12	No conflicts	Empty state	Banner and cards are replaced by the positive card alone	Continue remains the only action

PURPOSE

Cross-checks the generated list against the profile, and lets the user act without leaving the screen.

ENTERED FROM

SCR-05 · the alerts cell on SCR-04

EXITS TO

Replacement returns here · Continue opens SCR-05

DATA SOURCE

AllergyMatcher(profile.allergies, condition) over the item list

STATES

Conflicts found · no conflicts · replacement in progress

Shows the values every other screen reads, and is the only place they can be changed.



#	Element	Composable	Specification	Behaviour
1	Screen title	Text	headline/26sp Bold, 20dp gutter	No back — this is a nav root
2	Profile picture	Box + Text	168dp circle, primary.container, 8dp surface.card ring, centre-cropped	Initials shown when no image is set
3	User information	Column	Name in headline/26sp Bold, member-since in caption/14sp muted, centred	Name is editable through SCR-03
4	Information card	C-03	118dp, 14dp radius, 1dp outline, one row per field	Values are read directly from the profile row
5	Region and household	Row	Label caption/14sp muted left, value title/20sp Bold right, 1dp divider between	Changing either regenerates quantities on the next run
6	Allergy label	C-10	label/12sp caps, 24dp above the chip row	Static
7	Allergy list	FlowRow of C-09	32dp chips, danger.container fill, 8dp gap, wraps	Empty state reads None recorded
8	Health condition	C-09	32dp chip, warning.container fill	Exactly one; None is displayed rather than hidden
9	Edit profile	C-01	56dp, primary, 20dp gutters	Opens SCR-03 pre-filled and in edit mode
10	Log out	C-02 (danger)	52dp, pill, 2dp danger border, danger label	Confirmation dialog naming what is kept on the device
11	Bottom navigation	C-12	72dp, Profile selected	Fade between tabs
12	Persistence	ProfileDao	Single row; log out clears the session, not the grocery history	The dialog states this explicitly

PURPOSE

Shows the values every other screen reads, and is the only place they can be changed.

ENTERED FROM

Bottom nav · avatar on SCR-04

EXITS TO

SCR-03 in edit mode · SCR-02 on log out

DATA SOURCE

ProfileDao.observe() · aggregate counts

STATES

Default · editing · log-out confirmation